

FinSecure Payments Pvt. Ltd.

Real-Time Fraud Detection System — Project Brief
Confidential | Pune, India | Fintech / Digital Payments

Organization	FinSecure Payments Pvt. Ltd.
Location	Pune, India
Industry	Fintech / Digital Payments
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1. Background

FinSecure processes over **2.3 million transactions daily** across its mobile payment platform. Over the past 6 months, the company has observed a significant rise in fraudulent transactions — particularly card-not-present fraud and account takeover attempts — resulting in financial losses exceeding **₹4.2 crore per quarter**.

The current rule-based fraud detection system flags transactions based on static thresholds (e.g., transaction amount > ₹50,000). This approach is generating:

- High false positives — legitimate customers being blocked, causing churn
- High false negatives — sophisticated fraud slipping through undetected
- Manual review backlog — risk team overwhelmed with alerts

2. Client Requirement

"We need a system that can evaluate every incoming transaction and determine in under 200 milliseconds whether it is fraudulent or legitimate — before the payment is approved. The system must learn from patterns in historical transaction data and flag suspicious activity in real time without disrupting genuine customer experience."

— Head of Risk & Compliance, FinSecure Payments

3. Functional Requirements

- Accepts a single transaction payload via REST API
- Returns a fraud probability score (0.0 to 1.0) and a decision: approve / review / block
- Response time under 200ms
- Must handle peak load of 500 requests/second

4. Input Data Available

Feature	Description
transaction_id	Unique transaction identifier
user_id	Customer identifier
transaction_amount	Amount in INR
merchant_category	e.g. grocery, electronics, travel
transaction_hour	Hour of day (0–23)
day_of_week	0 = Monday, 6 = Sunday
device_type	mobile / web / POS
location_city	City of transaction
distance_from_home	Distance in km from registered address
is_new_device	1 = First time using this device, 0 = known device
failed_attempts_last_1hr	Failed login/payment attempts in last 1 hour
avg_transaction_amount_30d	Customer's 30-day average transaction amount
is_fraud	Target variable — 1 = fraud, 0 = legitimate

5. Expected API Contract

Request — POST /predict

```
POST /predict
{
  "transaction_amount": 45000,
  "merchant_category": "electronics",
  "transaction_hour": 2,
  "day_of_week": 6,
  "device_type": "mobile",
  "distance_from_home": 312.5,
  "is_new_device": 1,
  "failed_attempts_last_1hr": 3,
  "avg_transaction_amount_30d": 8200
}
```

Response

```
{
  "fraud_probability": 0.91,
  "decision": "block",
  "response_time_ms": 143
}
```

6. Success Criteria

Metric	Target
Precision	> 90%
Recall	> 85%

False Positive Rate	< 5%
API Response Time	< 200ms
System Uptime	99.9%

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